# Submission PHYS1-PS2-SUB02

Question PHYS1-PS2-Q01
work / setup:
1. v=v■+at=5+3(4)=17 m/s
crossed-out arithmetic: [smudged]
FINAL (maybe): v=17 m/s; displacement=44 m plus 1

Question PHYS1-PS2-Q02
work / setup:
1. N=mg=29.4 N; friction=μN=5.9 N
2. ΣF=19–5.9=13.1 N
crossed-out arithmetic: [smudged]
FINAL (maybe): a=4.37 m/s$^2$; friction opposes motion plus 1

Question PHYS1-PS2-Q03
work / setup:
1. mgh=½mv$^2$ ⇒ v=√(2gh)=√(2·9.8·6)
crossed-out arithmetic: [smudged]
FINAL (maybe): v=10.84 m/s plus 1

Question PHYS1-PS2-Q04
work / setup:
1. p_total,before = p_total,after
crossed-out arithmetic: [smudged]
FINAL (maybe): total momentum before equals after when external impulse is negligible plus 1

Question PHYS1-PS2-Q05
work / setup:
1. Impulse is area under F(t): J=FΔt=6(0.5)=3 N·s.
crossed-out arithmetic: [smudged]
FINAL (maybe): impulse=3 N·s; momentum changes by 3 kg·m/s plus 1

Question PHYS1-PS2-Q06
work / setup:
1. v=v■+at=4+2(5)=14 m/s
crossed-out arithmetic: [smudged]
FINAL (maybe): v=14 m/s; displacement=45 m plus 1

Question PHYS1-PS2-Q07
work / setup:
1. N=mg=19.6 N; friction=μN=3.9 N
2. ΣF=11–3.9=7.1 N
crossed-out arithmetic: [smudged]

FINAL (maybe): $a=3.54$ m/s$^2$; friction opposes motion plus 1

Question PHYS1-PS2-Q08
work / setup:
1. $mgh=\frac{1}{2}mv^2 \Rightarrow v=\sqrt{(2gh)}=\sqrt{(2\cdot 9.8\cdot 6)}$
crossed-out arithmetic: [smudged]
FINAL (maybe): $v=10.84$ m/s plus 1